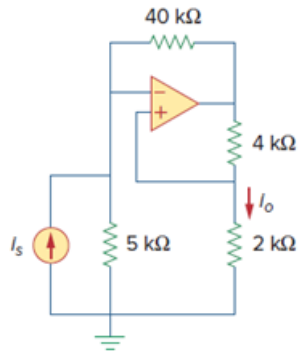


Problem

The op amp circuit in Fig. 5.107 is a *current amplifier*. Find the current gain i_o/i_s of the amplifier.

Figure 5.107:



Step-by-step solution

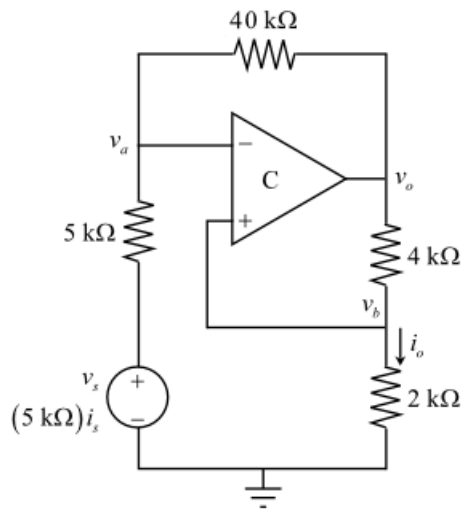
Step 1 of 5

Refer to Figure 5.107 in the textbook for current amplifier circuit.

Use the Ohm's law to transform the current source, i_s into a voltage source, v_s .

$$v_s = i_s (5 \text{ k}\Omega) \dots\dots (1)$$

The following is the modified circuit after transforming the current source into a voltage source:



Step 2 of 5

Apply voltage division method at node b to find the voltage v_b .

$$\begin{aligned} v_b &= \left(\frac{2 \text{ k}\Omega}{2 \text{ k}\Omega + 4 \text{ k}\Omega} \right) v_o \\ &= \left(\frac{2 \text{ k}\Omega}{6 \text{ k}\Omega} \right) v_o \\ &= \frac{v_o}{3} \end{aligned}$$

The voltage at the inverting and noninverting terminals of an ideal operational amplifier is same, since there is no current flowing into the input terminals of the operational amplifier. Therefore,

$$v_a = v_b$$

$$v_a = \frac{v_o}{3} \dots\dots (2)$$

Step 3 of 5

Use the Ohm's law to find the current, i_o .

$$\begin{aligned} i_o &= \frac{v_b}{2 \text{ k}\Omega} \\ &= \frac{\left(\frac{v_o}{3} \right)}{2 \text{ k}\Omega} \\ &= \frac{v_o}{6 \text{ k}\Omega} \end{aligned}$$

Write the expression for output voltage using this expression.

$$v_o = (6 \text{ k}\Omega) i_o \dots\dots (3)$$

Step 4 of 5

The current flow in to the input terminals of an ideal operational amplifier is zero.

Apply Kirchhoff's current law at node a .

$$\begin{aligned} \frac{v_s - v_a}{5 \text{ k}\Omega} &= \frac{v_a - v_o}{40 \text{ k}\Omega} \\ 40 \text{ k}\Omega (v_s - v_a) &= 5 \text{ k}\Omega (v_a - v_o) \\ (40 \text{ k}\Omega) v_s + (5 \text{ k}\Omega) v_o &= (5 \text{ k}\Omega) v_a + (40 \text{ k}\Omega) v_a \\ (5 \text{ k}\Omega) v_o &= (45 \text{ k}\Omega) v_a - (40 \text{ k}\Omega) v_s \end{aligned}$$

Substitute $(5 \text{ k}\Omega) i_o$ for v_s from equation (1), $\frac{v_o}{3}$ for v_a from equation (2), and $(6 \text{ k}\Omega) i_o$ for v_o from equation (3).

$$\begin{aligned} (5 \text{ k}\Omega)(6 \text{ k}\Omega) i_o &= (45 \text{ k}\Omega) \left(\frac{v_o}{3} \right) - (40 \text{ k}\Omega)(5 \text{ k}\Omega) i_o \\ (30 \text{ M}\Omega) i_o &= (45 \text{ k}\Omega) \left(\frac{6 \text{ k}\Omega}{3} \right) i_o - (200 \text{ M}\Omega) i_o \\ (30 \text{ M}\Omega) i_o &= (90 \text{ M}\Omega) i_o - (200 \text{ M}\Omega) i_o \\ (200 \text{ M}\Omega) i_o &= (90 \text{ M}\Omega) i_o - (30 \text{ M}\Omega) i_o \end{aligned}$$

Step 5 of 5

Simplify further to get an expression for $\frac{i_o}{i_s}$.

$$(200 \text{ M}\Omega)i_s = (60 \text{ M}\Omega)i_o$$

$$\frac{i_o}{i_s} = \frac{200 \text{ M}\Omega}{60 \text{ M}\Omega}$$
$$= 3.33$$

Thus, the current gain, $\frac{i_o}{i_s}$ of the current amplifier circuit is 3.33.